

1 ADJUSTMENT FACTORS

SAWN LUMBER · WHERE: NDS §4.3, TABLE 4.3.1

		ASD				ASD + LRFD							LRFD ONLY		
		LOAD DURATION	WET SERVICE	TEMPERATURE	BEAM STABILITY	SIZE	FLAT USE	INCISING	REPETITIVE MEMBER	COLUMN STABILITY	BUCKLING STIFFNESS	BEARING AREA	FORMAT CONV. K_F	RESISTANCE ϕ	TIME EFFECT
$F'_b = F_b$	×	C_D	C_M	C_t	C_L	C_F	C_{fu}	C_i	C_r	-	-	-	2.54	0.85	λ
$F'_t = F_t$	×	C_D	C_M	C_t	-	C_F	-	C_i	-	-	-	-	2.70	0.80	λ
$F'_v = F_v$	×	C_D	C_M	C_t	-	-	-	C_i	-	-	-	-	2.88	0.75	λ
$F'_c = F_c$	×	C_D	C_M	C_t	-	C_F	-	C_i	-	C_P	-	-	2.40	0.90	λ
$F'_{c\perp} = F_{c\perp}$	×	-	C_M	C_t	-	-	-	C_i	-	-	-	C_b	1.67	0.90	-
$E' = E$	×	-	C_M	C_t	-	-	-	C_i	-	-	-	-	-	-	-
$E'_{\min} = E_{\min}$	×	-	C_M	C_t	-	-	-	C_i	-	-	C_T	-	1.76	0.85	-

1.1 WET SERVICE FACTOR C_M

WHERE: NDS §4.3.3 (RULE) · VALUES: NDS SUPPLEMENT, IN THE ADJUSTMENT-FACTOR BLOCK PRINTED AFTER EACH DESIGN-VALUE TABLE (4A, 4B, 4C, 4D)

Dimension lumber with moisture content > 19% for an extended period: multiply design values by C_M below.

F_b	F_t	F_v	$F_{c\perp}$	F_c	$E, E_{\min}$
0.85*	1.0	0.97	0.67	0.8**	0.9

* $F_b C_F \leq 1150 \text{ psi} \rightarrow C_M = 1.0$ ** $F_c \leq 750 \text{ psi} \rightarrow C_M = 1.0$ 1.2 LOAD DURATION FACTOR C_D (ASD ONLY)

WHERE: NDS §4.3.2 POINTS TO §2.3.2 · VALUES: TABLE 2.3.2 (CHAPTER 2)

LOAD DURATION	C_D^*	TYPICAL DESIGN LOAD
Permanent	0.9	Dead
Ten years	1.0	Occupancy live
Two months	1.15	Snow
Seven days	1.25	Construction
Ten minutes	1.6	Wind / earthquake
Impact**	2.0	Impact

* C_D never applies to $E, E_{\min}, F_{c\perp}$ ** $C_D > 1.6$: not for pressure-treated (water-borne) or fire-retardant lumber
impact C_D : not for connections1.3 TEMPERATURE FACTOR C_t

WHERE: NDS §4.3.4 POINTS TO §2.3.3 · VALUES: TABLE 2.3.3 (CHAPTER 2)

Sustained exposure up to 150°F (see NDS Appendix C).

REF. DESIGN VALUES	MOISTURE*	$T \leq 100^\circ\text{F}$	$100 < T \leq 125$	$125 < T \leq 150$
$F_t, E, E_{\min}$	Wet or dry	1.0	0.9	0.9
$F_b, F_v, F_c, F_{c\perp}$	Dry	1.0	0.8	0.7
	Wet	1.0	0.7	0.5

* wet/dry service definitions: §4.1.4 sawn · 5.1.4 glulam · 7.1.4 I-joist
§8.1.4 SCL · 9.3.3 panels · 10.1.5 CLT

1.4 BEAM STABILITY FACTOR C_L

WHERE: NDS §4.3.5 POINTS TO §3.3.3 (CHAPTER 3) · EFFECTIVE LENGTH: TABLE 3.3.3

$C_L = 1.0$ WHEN

§3.3.3.1 $d \leq b$ (no lateral support required)

§3.3.3.2 rectangular sawn lumber laterally supported per §4.4.1

§3.3.3.3 compression edge braced full length + bearing ends held against rotation

$d > b$: lateral support required at bearing points (§3.3.3.4). ℓ_u = distance between bearing points, intermediate lateral supports, or cantilever length. ℓ_e from Table 3.3.3 (§3.3.3.5).

d member depth, in

b member breadth, in

ℓ_u laterally unsupported length, in

ℓ_e effective span length, in

TABLE 3.3.3 – LOADING	$\ell_u/d < 7$	$\ell_u/d \geq 7$
CANTILEVER ¹		
UDL	$1.33 \ell_u$	$0.90 \ell_u + 3d$
P at unsupported end	$1.87 \ell_u$	$1.44 \ell_u + 3d$
SINGLE SPAN ^{1,2}		
UDL	$2.06 \ell_u$	$1.63 \ell_u + 3d$
P at center, no int. support	$1.80 \ell_u$	$1.37 \ell_u + 3d$
P at center + support at center		$1.11 \ell_u$
2P at 1/3 pts + supports there		$1.68 \ell_u$
3P at 1/4 pts + supports there		$1.54 \ell_u$
4P at 1/5 pts + supports there		$1.68 \ell_u$
5P at 1/6 pts + supports there		$1.73 \ell_u$
6P at 1/7 pts + supports there		$1.78 \ell_u$
7P+ evenly spaced + supports there		$1.84 \ell_u$
Equal end moments		$1.84 \ell_u$

¹ loading not listed: $2.06 \ell_u$ ($\ell_u/d < 7$) · $1.63 \ell_u + 3d$ ($7 \leq \ell_u/d \leq 14.3$) · $1.84 \ell_u$ (> 14.3)

² multiple spans: table values or engineering analysis

1.5 SIZE FACTOR C_F

WHERE: NDS §4.3.6 · VALUES: NDS SUPPLEMENT TABLE 4A/4B ADJUSTMENT BLOCK

Visually graded dimension lumber 2–4" thick: multiply F_b , F_t , F_c (§4.3.6.1). Table 4D timbers with depth $> 12''$ (§4.3.6.2):

$$C_F = (12/d)^{1/9} \leq 1.0 \quad (1)$$

GRADES	WIDTH	F_b BY THICKNESS			
		2" & 3"	4"	F_t	F_c
Select Str.,	2", 3", 4"	1.5	1.5	1.5	1.15
No.1 & Btr.,	5"	1.4	1.4	1.4	1.1
No.1, No.2,	6"	1.3	1.3	1.3	1.1
No.3	8"	1.2	1.3	1.2	1.05
	10"	1.1	1.2	1.1	1.0
	12"	1.0	1.1	1.0	1.0
	14" & wider	0.9	1.0	0.9	0.9
Stud	2", 3", 4"	1.1	1.1	1.1	1.05
	5" & 6"	1.0	1.0	1.0	1.0
	8" & wider	use No.3 values + size factors			
Constr., Std.	2", 3", 4"	1.0	1.0	1.0	1.0
Utility	4"	1.0	1.0	1.0	1.0
	2" & 3"	0.4	–	0.4	0.6

1.6 FLAT USE FACTOR C_{fu}

WHERE: NDS §4.3.7 · VALUES: NDS SUPPLEMENT TABLE 4A/4B/4C/4F ADJUSTMENT BLOCK

Dimension lumber 2–4" thick loaded flatwise (load on wide face): F_b may also be multiplied by C_{fu} .

WIDTH (DEPTH)	THICKNESS (BREADTH)	
	2" & 3"	4"
2" & 3"	1.0	–
4"	1.1	1.0
5"	1.1	1.05
6"	1.15	1.05
8"	1.15	1.05
10" & wider	1.2	1.1

NOTE Table 4A shading: values based on 4" width (Constr./Std./Utility) ·
6" width (Stud) · 12" width (Select Str., No.1 & Btr, No.1–No.3)

1.7 INCISING FACTOR C_i

WHERE: NDS §4.3.8 · VALUES: TABLE 4.3.8 (MAIN SPEC, CHAPTER 4)

Dimension lumber incised parallel to grain: max depth 0.4", max length 3/8".

DESIGN VALUE	C_i
$E, E_{\min}$	0.95
F_b, F_t, F_c, F_v	0.80
$F_{c\perp}$	1.00

1.8 REPETITIVE MEMBER FACTOR C_r

WHERE: NDS §4.3.9 · SINGLE VALUE, NO TABLE

$C_r = 1.15$ on F_b , dimension lumber 2–4" thick (Tables 4A, 4B, 4C, 4F), when all of:

- 1 ≥ 3 members (joists, truss chords, rafters, studs, planks, decking)
- 2 in contact or spaced $\leq 24"$ o.c.
- 3 joined by floor/roof/other load-distributing element

NOTE Table 4E visually graded decking: $C_r = 1.15$ already baked in

1.9 COLUMN STABILITY FACTOR C_P

WHERE: NDS §4.3.10 POINTS TO §3.7 (COMPUTED, NO TABLE)

Applies to F_c .

1.10 BUCKLING STIFFNESS FACTOR C_T

WHERE: NDS §4.3.11 POINTS TO §4.4.2

Applies to $E_{\min}$ only.

1.11 BEARING AREA FACTOR C_b

WHERE: NDS §4.3.12 POINTS TO §3.10.4

Applies to $F_{c\perp}$ (permitted increase).